

MASENO 2025 KCSE PREDICTION



CONFIDENTIAL EXAM

121/1

MATHEMATICS

PAPER 1

TIME: 2½ HOURS



NAME.....

SCHOOL..... SIGN.....

INDEX NO..... ADM NO.....

Instructions to candidates

1. Write your name, index number, school and stream in the spaces provided above.
2. This paper contains two sections: **Section I** and **Section II**. Attempt **all** the questions in **section I** and **strictly any five questions from Section II**.
3. All answers and working must be written on the question paper in the spaces provided below each question.
4. Show all the steps in your calculations, giving your answers at each stage in the spaces below each question.
5. Marks may be given for correct working even if the answer is wrong.
6. Non-programmable silent electronic calculators and **KNEC** mathematical tables may be used, except where stated otherwise.
7. Answer the questions in English

FOR OFFICIAL USE ONLY:

SECTION I

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	Total

SECTION II

17	18	19	20	21	22	23	24	Total

GRAND TOTAL

SECTION I: (50 MARKS)

Attempt all questions in the spaces provided

1. An irregular 6 sided polygon has two of its interior angles equal to $2x$ each, three angles equal to x each and one side equal to 20°

Calculate the value of x

(3marks)

2. The diagonals of a parallelogram are 20cm and 28.8cm. The acute angle between the diagonals is 62° . Calculate the area of the parallelogram.

(3marks)

3. A mobile phone seller gets a commission of shs 250 on every mobile phone that he sells. In a given month he got 33,000 shillings.

a. How many phones did he sell that month

(1mark)

b. If this commission is 2% what is the sale price of each mobile phone. **(2marks)**

4. Given that $\sin x = 3/4$, without using tables or calculators, find.

a. $\cos x^\circ$ **(2marks)**

b. $\tan (90-x)^\circ$ **(1mark)**

5. Solve the simultaneous equation

$$2\log y = \log 2 + \log t$$

$$2^y = 4^t$$

(4marks)

6. Evaluate without using tables or a calculator

$$100^{-1.5} \times 32^{0.2}$$

(3marks)

7. The angle subtended by the major arc at the center of the circle O is twice the angle subtended by the minor arc at the center. If the radius of the circle is 3.5cm. Find the length of the minor arc.

(3marks)

8. Two trains T_1 and T_2 travelling in the opposite directions on a parallel tracks are just beginning to pass one another. Train T_1 is 72m long and is travelling at 108km/hr. Train T_2 is 78m long and is travelling at 72km/hr. Find the time in seconds the two trains take to completely pass one another

(3marks)

9. The angle of depression of a chick from the hawk on top of a vertical tree 8m tall is 28° . On seeing the hawk, the chick moves directly towards the base of the tree to a point P such that the angle of elevation of the hawk from P is 32° . Calculate the distance moved by the chick.

(4marks)

10. Find a 2×2 matrix m such that

(3marks)

$$\begin{pmatrix} 1 & 2 \\ 2 & -1 \end{pmatrix}^2 m \begin{pmatrix} 1 & 7 \\ 3 & -2 \end{pmatrix} = \begin{pmatrix} 4 & -6 \\ 2 & 0 \end{pmatrix}$$

11. Given that $a=1.2$, $b=0.02$ and $c=0.2$ Express $ac + b$ in the form $\frac{m}{n}$ where m and n are integers

(3mark)

12. The line passing through the point $P(-1,3w)$ and $Q(w,3)$ is parallel to another line whose equation is $2y - 3x = 9$.

Write down the co-ordinates of P and Q

(3marks)

13. A certain volume of solution has a mass of 2.2kg with density of 0.8g/cm^3 . Calculate the volume of the solution in liters

(3marks)

14. The graph below shows frequency densities for the masses of same 200 students selected from a class. Use it to answer the questions that follows.



(a) Complete the frequency distribution table below.

(2 marks)

Mass in kg					
frequency					

(b) State the modal frequency

(1mark)

15. Calculate the area bounded by the curve $y=x^2$, the line $y= -1$ and the lines $x=0$ and $x=3$ using trapezoidal rule with 7 ordinates (correct to 3 d p) (3marks)

16. Mutai imports rice from United States at the initial cost of 500 US dollars per tonnes. He then pays 20% of this amount as shopping costs and 10% of the same amount as custom duty. When the rice reaches Mombasa he has to pay 6% of the initial cost to transport it to Nairobi. Given that on the day of this transaction the exchange rate was 1 us dollar =kshs 76.60. calculate the total cost of importing one

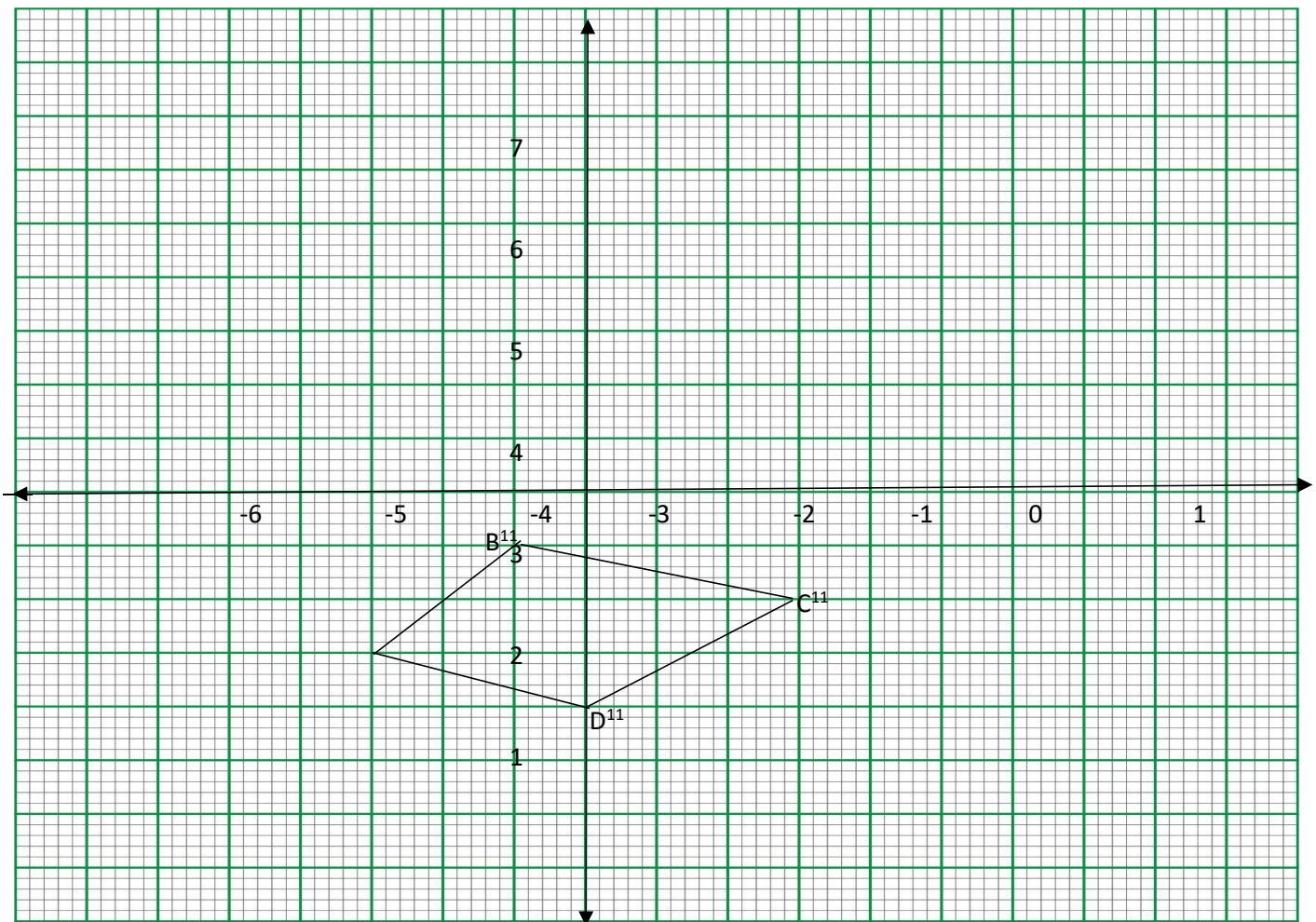
For Marking Schemes Contact 0746 222 000 / 0742 999 000

tonne of rice up to Nairobi in Kenya shillings. (3mks)

SECTION II (50 MARKS)

Answer only 5 questions from this section.

17. (a) On the Cartesian plane given below, draw the quadrilateral ABCD with vertices A(6,6), B(2,2)



C(4,-6) and D(8,0).

(1mrk)

- a. Draw the image $A^1B^1C^1D^1$ of ABCD under an enlargement scale factor $\frac{1}{2}$, center origin. State the coordinates of $A^1B^1C^1D^1$

(3mrks)

- b. Describe the transformation that maps $A^1B^1C^1D^1$ onto the given image $A^{11}B^{11}C^{11}D^{11}$ **(2mark)**

- c. Rotate $A^{11}B^{11}C^{11}D^{11}$ about center $(-2,-1)$ through a positive quarter turn to get $A^{111}B^{111}C^{111}D^{111}$. State the co-ordinates of $A^{111}B^{111}C^{111}D^{111}$ **(3marks)**

- d. State a pair of quadrilaterals that are oppositely congruent.

(1mark)

- 18.** On a certain day, Mwema bought plates worth Kshs 1200. On another day, Mrs. Mwema spent the same amount of money but bought the plates at a discount of 20% per plate.
- a.** If Mwema bought a plate at Kshs x , write down a simplified expression for the total number of plates bought by the two people **(3marks)**
- b.** If Mrs. Mwema bought 6 plates more than her husband, find how much each spent on a plate. **(5marks)**
- c.** Find the total number of plates bought by the family **(2marks)**

- 19. (a)** Using a ruler and a pair of compasses only, construct triangle ABC such that lines $AB=5.5\text{cm}$, $BC=4.8\text{cm}$ and $AC=6.8\text{cm}$, Construct circle passing through vertices A, B and C
- (6 marks)**

- a.** Measure the radius of the circle **(1mark)**
- b.** Measure the angle subtended at the center of the circle by chord AC **(1mark)**
- c.** Hence calculate the area of the triangle AOC **(2marks)**

20. A particle P moves in a straight line such that t seconds after passing a fixed point Q, its velocity is given by the equation $2t^2 - 10t + 12$

Find

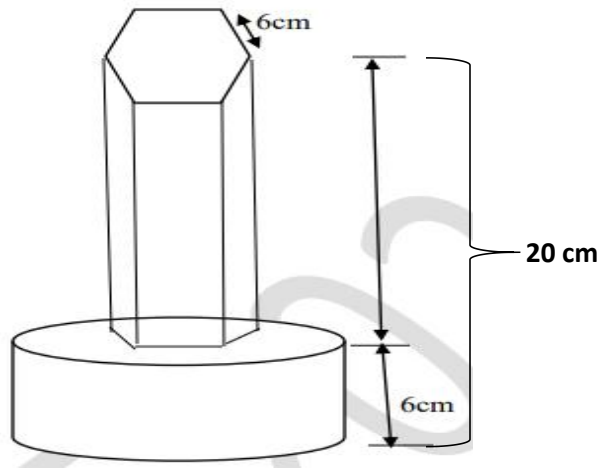
a. The values of t when P is instantaneously at rest. **(2marks)**

b. An expression for the distance moved by P after t seconds. **(2marks)**

c. The total distance travelled by P in the first 3 seconds after passing point Q **(2marks)**

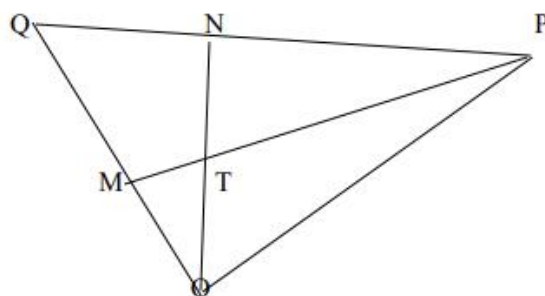
d. The distance of P from Q when acceleration is zero. **(4marks)**

- 21.** The figure below shows a model of a pillar to be constructed at the Canterbury. The model consists of a circular base of diameter 21cm and a uniform hexagon stand of sides 6cm and height 20cm.



- a.** Calculate the cross-sectional area of the hexagon to 2dp. **(3marks)**
- b.** Calculate the total volume of the model to 2dp. **(3marks)**
- c.** If the height of the red pillar is 52m and the constructor uses two bags of cement for every 500m^3 of the construction. Calculate the least number of bags of cement required **(4marks)**

22. The diagram below shows a triangle OPQ in which QN: NP=1:2, OT: TN=3:2 and M is the mid-point of OQ.



- (a) Given that $\vec{OP} = \mathbf{p}$ and $\vec{OQ} = \mathbf{q}$. Express the following vectors in terms of $\mathbf{p}$ and $\mathbf{q}$

(i) $\vec{PQ}$

(1mark)

(ii) $\vec{ON}$

(2marks)

(iii) $\vec{PT}$

(2marks)

(iv) $\vec{PM}$

(1mark)

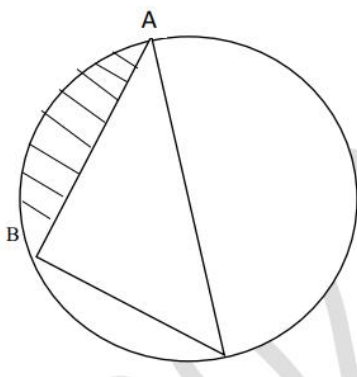
- (b) (i) Show that points P, T and M are collinear

(3marks)

- (ii) Determine the ratio MT : TP

(1mark)

- 23.** The diagram below shows a triangle ABC Circumscribed in a circle with $AB=12\text{cm}$ $BC=12\text{cm}$, $BC=15\text{cm}$ and $AC=14\text{cm}$



Calculate to 4 significant figures

- (a)** The angle ACB

(3marks)

- (b)** The radius of the circle

(3marks)

- (c)** The area of the shaded region

(4marks)

24. If Nick gives a quarter of the money he owns to Tom, Tom will have twice as much as Nick. If Tom gives q shillings to Nick, then Nick will have thrice as much as Tom. Taking the initial amount owned by Nick and Tom to be x and y respectively;

a. Express y and q in terms of x

(7marks)

b. Given that Nick's initial amount was Kshs 40,000. calculate the value of q

(1mark)

c. The initial amount owned by Tom

(2marks)

THIS IS THE LAST PRINTED PAGE